Zepto SQL Analytics Project

```sql
DROP TABLE IF EXISTS zepto;

CREATE TABLE IF NOT EXISTS zepto(
      sku_id SERIAL PRIMARY KEY,
    Category VARCHAR(100),
    name VARCHAR(100) NOT NULL,
    mrp NUMERIC(8,2),
    discountPercent NUMERIC(5,2),
    availableQuantity INT,
    discountedSellingPrice NUMERIC(8,2),
    weightInGms INT,
    outOfStock BOOLEAN,
    quantity INT
);

SELECT * FROM zepto;

--IMPORT CSV FILE
COPY
zepto(Category,name,mrp,discountPercent,availableQuantity,discountedSellingPrice,weightIn
Gms,outOfStock,quantity)
FROM 'D:\Data Analyst\SQL\Projects\Zepto\zepto_v2.csv'
DELIMITER ','
CSV HEADER;

-- Find total Count of Product
SELECT COUNT(*) FROM zepto;
```

	count bigint
1	3731

```sql
--Null values
SELECT * FROM zepto
WHERE name IS NULL
OR
category IS NULL
OR
mrp IS NULL
OR
discountPercent IS NULL
OR
discountedSellingPrice IS NULL
OR
weightInGms IS NULL
OR
availableQuantity IS NULL
OR
outOfStock IS NULL
OR
quantity IS NULL;
```

	sku_id [PK] integer	category character varying (100)	name character varying (100)	mrp numeric (8,2)	discountpercent numeric (5,2)	availablequantity integer	discountedsellingprice numeric (8,2)	weightingms integer	outofstock boolean	quantity integer

Total rows: 0 | Query complete 00:00:00.161 | CRLF | Ln 48, Col 31

```
--Different product categories
SELECT DISTINCT category
FROM zepto
ORDER BY category;
```

	category character varying (100)
1	Beverages
2	Biscuits
3	Chocolates & Candies
4	Cooking Essentials
5	Dairy, Bread & Batter
6	Fruits & Vegetables
7	Health & Hygiene

Total rows: 14 | Query comple

```
--Products in stock vs out of stock
SELECT outOfStock, COUNT(sku_id)
FROM zepto
GROUP BY outOfStock;
```

	outofstock boolean	count bigint
1	false	3278
2	true	453

```
--Product names present multiple times
SELECT name, COUNT(sku_id) AS "Number of SKUs"
FROM zepto
GROUP BY name
HAVING count(sku_id) > 1
ORDER BY count(sku_id) DESC;
```

	name character varying (100)	Number of SKUs bigint
1	Arden Eggs White	10
2	Amul Delicious Fat Spread - Cholesterol Free	10
3	Mother's Recipe Tamarind Paste	10
4	Quaker Oats	10
5	Sunfeast Yippee! Pasta Treat - Sour Cream Onion	10
6	Saffola Veggie Twist Masala Oats	10
7	Kellogg's Real Almond & Honey Corn Flakes	9

Total rows: 1214 | Query complete 00:00:00.117

```sql
--Data Cleaning
--Products with price = 0
SELECT * FROM zepto
WHERE mrp = 0 OR discountedSellingPrice = 0;

DELETE FROM zepto
WHERE mrp = 0;
```

```
DELETE 1

Query returned successfully in 281 msec.
```

```sql
--Convert paise to rupees
UPDATE zepto
SET mrp = mrp / 100.0,
discountedSellingPrice = discountedSellingPrice / 100.0;

SELECT mrp, discountedSellingPrice FROM zepto;
```

	mrp numeric (8,2)	discountedsellingprice numeric (8,2)
1	0.00	0.00
2	0.01	0.01
3	0.01	0.01
4	0.00	0.00
5	0.00	0.00

Total rows: 3731 Query complete 00:00:00.1

```sql
-- Data Analysis
-- Find all products with discount more than 15%
SELECT name, discountPercent
From Zepto
Where discountPercent>15;
```

	name character varying (100)	discountpercent numeric (5,2)
1	Zucchini Yellow	16.00
2	Apple Kinnaur	16.00
3	Parsley	17.00
4	Ridge Gourd	16.00
5	Mango Raw	16.00
6	Banana Nendran	16.00
7	Pumpkin Cut	16.00

Total rows: 475 Query complete 00:00:00.244

```sql
-- Show products sorted by hight MRP in descending order
SELECT DISTINCT name,mrp
FROM zepto
ORDER BY mrp DESC;
```

	name character varying (100)	mrp numeric (8,2)
1	Borges Extra Light Olive Oil Bottle	0.26
2	Pampers Pants - Large	0.15
3	Praakritik Natural Desi Gir Cow A2 Ghee	0.15
4	Dhara Kachi Ghani Mustard Oil Jar	0.13
5	Saffola Gold (Jar)	0.12
6	Dhara Health Refined Sun Flower Oil Jar	0.12
7	Pampers Premium Care Pants - Large	0.11

Total rows: 1758 Query complete 00:00:00.121

```sql
-- Show count products in each category
SELECT DISTINCT category, COUNT(name) AS Count_Of_Products
FROM zepto
GROUP BY name,category;
```

	category character varying (100)	count_of_products bigint
1	Biscuits	2
2	Chocolates & Candies	3
3	Fruits & Vegetables	2
4	Dairy, Bread & Batter	1
5	Cooking Essentials	2
6	Health & Hygiene	1
7	Packaged Food	2

Total rows: 37 Query complete 00:00:00.157

```sql
-- Find average of discount price per category
SELECT category, ROUND(AVG(discountedSellingPrice),2) AS Avg_Discount_Price
FROM zepto
GROUP BY category,discountedSellingPrice;
```

	category character varying (100)	avg_discount_price numeric
1	Home & Cleaning	0.07
2	Cooking Essentials	0.11
3	Personal Care	0.08
4	Beverages	0.00
5	Biscuits	0.01
6	Packaged Food	0.02
7	Cooking Essentials	0.01

Total rows: 103 Query complete 00:00:00.156

```sql
--Find top 5 products with highest discount amount.
SELECT DISTINCT name, (mrp - discountedSellingPrice) AS discount_price
FROM zepto
ORDER BY discount_price DESC
LIMIT 5;
```

	name character varying (100)	discount_price numeric
1	Borges Extra Light Olive Oil Bottle	0.12
2	Pampers Pants - Large	0.07
3	Pampers Active Baby Diapers New Born Extra Small	0.03
4	Kelloggs Chocos - Moons & Stars	0.03
5	RRO Mozzarella Block Cheese	0.02

```sql
--Find products where discount is above average.
SELECT name, discountPercent
FROM zepto
WHERE discountPercent > (SELECT AVG(discountPercent) FROM zepto);
```

	name character varying (100)	discountpercent numeric (5,2)
1	Zucchini Yellow	16.00
2	Apple Kinnaur	16.00
3	Pineapple	15.00
4	Basil	15.00
5	Parsley	17.00
6	Ridge Gourd	16.00
7	Papaya Cut	15.00

Total rows: 1550 Query complete 00:00:00.172

```sql
-- Find out of stock products.
SELECT name,outOfStock
FROM zepto
WHERE outOfStock='true';
```

	name character varying (100)	outofstock boolean
1	Carrot	true
2	Beetroot	true
3	Moong Sprouts	true
4	Safal Frozen Mixed Vegetables	true
5	Safal Frozen Corn	true
6	Safal Frozen - Sweet Corn	true
7	Madhur Sugar	true

Total rows: 453 Query complete 00:00:00.137

```sql
--Categorize product based on discount.
SELECT name,discountPercent,
  CASE
    WHEN discountPercent>=17 THEN 'High'
    WHEN discountPercent BETWEEN 14 AND 16 THEN 'Medium'
    ELSE 'Low'
  END AS discount_category
FROM zepto;
```

	name character varying (100)	discountpercent numeric (5,2)	discount_category text
1	Zucchini Yellow	16.00	Medium
2	Apple Kinnaur	16.00	Medium
3	Pineapple	15.00	Medium
4	Basil	15.00	Medium
5	Parsley	17.00	High
6	Ridge Gourd	16.00	Medium
7	Papaya Cut	15.00	Medium

Total rows: 3731 Query complete 00:00:00.142

```
-- Find category with highest average MRP
SELECT category, ROUND(AVG(mrp),2) AS avg_mrp
FROM zepto
GROUP BY category
ORDER BY avg_mrp DESC
LIMIT 1;
```

	category character varying (100)	avg_mrp numeric
1	Ice Cream & Desserts	0.02

```
--Rank product by price within each category
SELECT category, name, mrp,
  RANK() OVER(PARTITION BY category ORDER BY MRP DESC ) AS Products_rank
FROM zepto;
```

	category character varying (100)	name character varying (100)	mrp numeric (8,2)	products_rank bigint
1	Beverages	RRO Mascarpone Cheese	0.04	1
2	Beverages	Amulya Dairy Whitener	0.04	1
3	Beverages	Amul Cheese Slices	0.04	1
4	Beverages	RRO Burrata Cheese	0.03	4
5	Beverages	RRO Goat Milk Fetta Cheee	0.03	4
6	Beverages	RRO Fresh Ricotta	0.03	4
7	Beverages	RRO Sour Cream	0.03	4

Total rows: 3731 Query complete 00:00:00.232

```
-- Find second highest priced product.
SELECT name, mrp
FROM zepto
ORDER BY mrp DESC
LIMIT 1 OFFSET 1;
```

	name character varying (100)	mrp numeric (8,2)
1	Borges Extra Light Olive Oil Bottle	0.26

```sql
-- Find product with low stock but high price
SELECT name, availableQuantity,mrp
FROM zepto
WHERE availableQuantity < 5 AND mrp>10000;

-- Show total inventory value
SELECT SUM(discountedSellingPrice * availableQuantity) AS total_value
FROM zepto;
```

	total_value numeric
1	224.02

```sql
-- Find total number of products available in stock
SELECT SUM(availableQuantity) AS total_stock
FROM zepto
```

	total_stock bigint
1	14959

```sql
-- Find the least discounted product
SELECT name, discountedSellingPrice
FROM zepto
ORDER BY discountedSellingPrice ASC
LIMIT 1;
```

	name character varying (100)	discountedsellingprice numeric (8,2)
1	Zucchini Yellow	0.00

```sql
-- Find products where selling price is less than 1200
SELECT name
FROM zepto
WHERE discountedSellingPrice< 1200;
```

	name character varying (100)
1	Zucchini Yellow
2	Apple Kinnaur
3	Pineapple
4	Basil
5	Parsley
6	Ridge Gourd
7	Papaya Cut

Total rows: 3731 Query complete 00:00:00.1

```sql
-- Find products where mrp is more than average mrp
SELECT name, mrp
FROM zepto
WHERE mrp > (SELECT AVG(mrp) FROM zepto);
```

	name character varying (100)	mrp numeric (8,2)
1	Avocado Indian Premium	0.02
2	Safal Frozen - Sweet Corn	0.02
3	Fortune Sunlite Refined Sunflower (P...	0.02
4	Aashirvaad Superior MP Atta	0.03
5	Amul Pure Ghee (Pouch)	0.03
6	Saffola Gold Oil (Pouch)	0.02
7	Fortune Sunlite Refined Sunflower (J...	0.09

```sql
-- Find product with highest available quantity
SELECT name, availableQuantity
FROM zepto
ORDER BY availableQuantity DESC;
```

	name character varying (100)	availablequantity integer
1	Johnson's Baby Powder	6
2	Apple Kinnaur	6
3	Banana Nendran	6
4	Banana Red	6
5	Radish White	6
6	Maggi 2 Minute Masala Noodles	6
7	iD Idli & Dosa Batter	6

Total rows: 3731 Query complete 00:00:00.152

```sql
--Find top 3 categories with highest average discount
SELECT category,ROUND(AVG(discountPercent),2) AS avg_discount
FROM zepto
GROUP BY category
ORDER BY avg_discount DESC
LIMIT 3;
```

	category character varying (100)	avg_discount numeric
1	Fruits & Vegetables	15.46
2	Meats, Fish & Eggs	11.03
3	Ice Cream & Desserts	8.32

```sql
--Find the price per gram for products above 100g and sort by best value.
SELECT DISTINCT name, weightInGms, discountedSellingPrice,
ROUND(discountedSellingPrice/weightInGms,2) AS price_per_gram
FROM zepto
WHERE weightInGms >= 100
ORDER BY price_per_gram;
```

	name character varying (100)	weightingms integer	discountedsellingprice numeric (8,2)	price_per_gram numeric
1	Johnson's Baby Oil	100	0.01	0.00
2	Britannia Vita Marie Gold Biscuit	300	0.00	0.00
3	Kelloggs Chocos - Moons & Stars	1200	0.04	0.00
4	Pillsbury Atta With Multigrains Wheat...	5000	0.02	0.00
5	Amul Cheese Spread Spicy Garlic	200	0.01	0.00
6	Cadbury Oreo Strawberry Bicuits	120	0.00	0.00
7	Head & Shoulders 2-in-1 Cool Menth...	340	0.04	0.00

Total rows: 1387 Query complete 00:00:00.292

```sql
-- Group the products into categories like Low, Medium, Bulk.
SELECT DISTINCT name, weightInGms,
CASE WHEN weightInGms < 1000 THEN 'Low'
  WHEN weightInGms < 5000 THEN 'Medium'
  ELSE 'Bulk'
  END AS weight_category
FROM zepto;
```

	name character varying (100)	weightingms integer	weight_category text
1	Popular Essentials Poppy Seeds (Ga...	100	Low
2	Origami Kitchen Towels 2 In 1 Quilte...	116	Low
3	Johnson's Baby Top To Toe Bath Bod...	210	Low
4	Harpic Hygienic Toilet Cleaner Rim Bl...	26	Low
5	Lizol Disinfectant Surface & Floor Cle...	500	Low
6	Sapota	348	Low
7	Britannia Treart Creme Wafers Choco...	60	Low

Total rows: 1782 Query complete 00:00:00.192

```sql
--What is the Total Inventory Weight Per Category
SELECT category,
SUM(weightInGms * availableQuantity) AS total_weight
FROM zepto
GROUP BY category
ORDER BY total_weight;
```

	category character varying (100)	total_weight bigint
1	Meats, Fish & Eggs	48016
2	Biscuits	84431
3	Fruits & Vegetables	91794
4	Health & Hygiene	142904
5	Dairy, Bread & Batter	143735
6	Beverages	143735
7	Paan Corner	348187

Total rows: 14 Query complete 00:00:00.186